MAT 221 - CALCULUS I
Lesson 18: - Area of Surfaces of Revolution

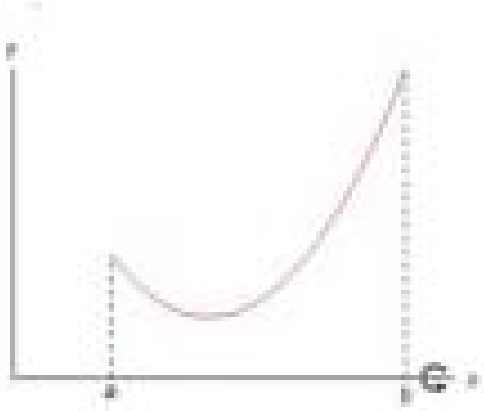
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Surface Area

- A **surface of revolution** is the surface that you get when you rotate a two dimensional curve around a specific axis.
- The image below shows a function f(x) over a closed interval [a, b], and the surface of revolution you get when you rotate it around the x axis





- If the function $f(x) \geqslant 0$ is continuously differentiable on $[a, b]$, the **area of the surface** generated by revolving the graph of $y = f(x)$ about the $x-$axis is

$$S = \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \qquad (1)$$

- If the function $g(y) \geqslant 0$ is continuously differentiable on $[c, d]$, the **area of the surface** generated by revolving the graph of $x = g(y)$ about the $y-$axis is

$$S = \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \tag{2}$$

- If the curve is **parametric**, that is $y = g(\theta)$ and $x = f(\theta)$ between $\theta_1 \leqslant \theta \leqslant \theta_2$, the **area of the surface** generated by revolving the graph about the $x-$axis is given by

$$S = \int_{\theta_1}^{\theta_2} 2\pi y \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta \tag{3}$$

- If the curve is **parametric**, that is $y = g(\theta)$ and $x = f(\theta)$ between $\theta_1 \leqslant \theta \leqslant \theta_2$, the **area of the surface** generated by revolving the graph about the $y-$axis is given by

$$S = \int_{\theta_1}^{\theta_2} 2\pi x \sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2}\, d\theta \tag{4}$$

Example 1

Determine the surface area of the solid obtained by rotating $y = \sqrt{9 - x^2}$, $-2 \leqslant x \leqslant 2$ about the $x-$axis.

Solution

Lets first get the derivative

$$\frac{dy}{dx} = \frac{1}{2}(9 - x^2)^{-\frac{1}{2}}(-2x) = -\frac{x}{(9 - x^2)^{\frac{1}{2}}}$$

Taking care of the root we get

$$\sqrt{1 + \left(\frac{dy}{dx}\right)^2} = \sqrt{1 + \frac{x^2}{9 - x^2}} = \sqrt{\frac{9}{9 - x^2}} = \frac{3}{\sqrt{9 - x^2}}$$

The surface area is then,

$$\begin{aligned} S &= \int_a^b 2\pi y \sqrt{1 + \left(\frac{dy}{dx}\right)^2}\, dx \\ &= \int_{-2}^{2} 2\pi \sqrt{9 - x^2} \frac{3}{\sqrt{9 - x^2}}\, dx \\ &= \int_{-2}^{2} 6\pi\, dx = 24\pi. \end{aligned}$$

Example 2

Determine the surface area of the solid obtained by rotating $y = \sqrt[3]{x}$, $1 \leqslant y \leqslant 2$ about the $y-$axis.

Solution

Solving for x we get

$$x = y^3, \quad \frac{dx}{dy} = 3y^2 \Rightarrow \sqrt{1 + \left(\frac{dx}{dy}\right)^2} = \sqrt{1 + 9y^4}$$

The surface area is then

$$\begin{aligned} S &= \int_c^d 2\pi x \sqrt{1 + \left(\frac{dx}{dy}\right)^2}\, dy \\ &= \int_1^2 2\pi y^3 \sqrt{1 + 9y^4}\, dy, \qquad \text{letting} \quad u = 1 + 9y^4 \\ &= \frac{\pi}{18} \int_{10}^{145} u^{\frac{1}{2}}\, du \\ &= \frac{\pi}{27}(145^{\frac{3}{2}} - 10^{\frac{3}{2}}) \\ &= 199.5 \end{aligned}$$

Example 3

Find the area generated when the curve $x = \theta - \sin\theta$, $y = 1 - \cos\theta$ between $\theta = 0$ and $\theta = \pi$, rotates about the $x-$axis.

Solution

$\frac{dx}{d\theta} = 1 - \cos\theta, \qquad \frac{dy}{d\theta} = \sin\theta$

$$\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2 = (1-\cos\theta)^2 + \sin^2\theta = 1 + \cos^2\theta + \sin^2\theta - 2\cos\theta$$

$$= 2(1-\cos\theta) = 4\sin^2\frac{\theta}{2}, \quad \because 1 - \cos\theta = 2\sin^2\frac{\theta}{2}$$

then

$$y\sqrt{\left(\frac{dx}{d\theta}\right)^2 + \left(\frac{dy}{d\theta}\right)^2} = (1-\cos\theta)\left(2\sin\frac{\theta}{2}\right) = \left(2\sin^2\frac{\theta}{2}\right)\left(2\sin\frac{\theta}{2}\right)$$

$$= 4\left(1-\cos^2\frac{\theta}{2}\right)\sin\frac{\theta}{2} = 4\left(\sin\frac{\theta}{2} - \cos^2\frac{\theta}{2}\sin\frac{\theta}{2}\right)$$

$$S = 8\pi\int_0^{\pi}\left(\sin\frac{\theta}{2} - \cos^2\frac{\theta}{2}\sin\frac{\theta}{2}\right)d\theta = 8\pi\left[-2\cos\frac{\theta}{2} + \frac{2}{3}\cos^3\frac{\theta}{2}\right]_0^{\pi} = \frac{32\pi}{3}$$